

Conflict Frame Detection at Scale: An Iterative Human-in-the-Loop LLM Annotation Pipeline Using Chilean Political News as a Case Study (2018-2026)

1. Introduction

Social media has become the primary source through which citizens inform themselves about politics. As a result, traditional news outlets had to reconfigure their entire business strategy. While traditional media logics used to be centered on circulation and editorial identity, social media ecosystems demand content success to be measured by its engagement. Recent studies show that more clicks and shares are generated by coverage that triggers a strong emotional response, which is why negative (Robertson et al., 2023) and sensationalist content (Khawar et al., 2024) have both been identified as effective editorial strategies to increase engagement.

While engagement-based algorithms amplify divisive and polarizing content (Milli et al., 2025; Jia et al., 2025), users, who are drawn in by this content, disproportionately share material that expresses hostility toward political opponents (Rathje et al., 2021). This is particularly relevant since polarization has been linked to directly undermining citizens' commitment to democratic norms (Carlin & Love, 2023).

Erving Goffman's (1974) framing analysis has emerged as a key framework to examine how editorial choices shape political information, and to systematically measure the impact of different frames on citizens' perception of reality and belief formation. One of the framings that have been largely studied in the political communications field is the conflict frame. Its academic relevance derives from its high intercoder reliability and its widely perceived presence in political coverage around the world (Semetko & Valkenburg, 2000). Moreover, research has shown that sustained exposure to it leads to political cynicism and disengagement (Cappella & Jamieson, 1997), further reinforcing the democratic consequences described above.

Latin American media ecosystems are especially relevant for studying conflict framing, given the rising political polarization (Sarsfield, Moncagatta & Roberts, 2024) and high media ownership concentration across the continent (Becerra & Mastrini, 2019). Chile stands out as a case study due to a significant number of recent events with sociological and political relevance. Studies have shown that the way the media portrays content has impacted citizens' perception in the country, particularly on issues such as crime (Scherman et al., 2022) and migration (Ajzenman, Dominguez & Undurraga, 2023). Over the past decade, small-sample

studies have shown that conflict is the most predominant frame in the country's political coverage well above the other frames measured (Gronemeyer & Porath, 2017a). Despite its relevance, the conflict frame has not been analyzed on a large-scale, in part for the methodological constraints that such analysis implies.

While theoretical frameworks for operationalizing framing are widely refined, the requirement of human annotators to manually identify their presence in each text has made large-scale analysis impossible until now. The recent introduction of Large Language Models (LLMs) as annotators has removed this barrier, at least partially. Recent studies on English datasets have shown that, when the task is well-defined and clearly specified, these models are able to annotate headlines or tweets with results comparable to human intercoder agreement levels (Gilardi et al., 2023; Pangakis & Wolken, 2024).

Now that large-scale annotation of short English texts can be automated for frame detection, the approach should be tested for other languages and longer text formats. The present study uses an LLM to annotate a large dataset in Spanish that covers Chilean political news from 2018 to 2026. It develops and tests an automated annotation pipeline to examine conflict framing patterns in Chilean news media on a large scale. Its findings indicate that the annotation pipeline performs well enough at the aggregate level to support the analysis proposed in this study, even though individual sub-dimensions do not meet this threshold on their own. Built entirely on open-source components, several parts of this pipeline are directly reusable for future research. Yet, reaching this level of performance required an iterative, human-in-the-loop process, which includes curating locally applicable examples and applying contextual, interpretive judgment.

Applied to Chilean political coverage, the results reveal that conflict framing in the country is characterized by three consistent patterns. First, Chilean political journalism tends to cover institutions and processes primarily through the individual or collective actors who lead them. This structure provides more textual space for conflict to be articulated, without this necessarily reflecting a higher underlying level of political conflict. Second, while outlets broadly agree on which topics are conflictive, they diverge substantially in how intensely they apply the frame, with parliamentary politics standing out as the only domain with near-uniform intensity. Third, the conflict frame does not track discrete politically relevant events consistently. Major elections and institutional transitions do not reliably generate comparable elevation. Instead, it follows a recurring seasonal pattern, with activation

consistently lower in February and higher in July and August across every year in the corpus.

2. Literature Review

2.1. Framing Theory

Framing theory proposes that a particular information can be made more noticeable to audiences by selecting certain aspects of it and making them more salient (Entman, 1993). It offers a way of conceptualizing the complex power dynamics that are present in the way that a text is communicated. By highlighting specific aspects, the communicator can shape the meaning of an issue in four ways: how problems are defined, how causes are diagnosed, how moral judgments are made, and what remedies are recommended.

The introduction of this theory was significant in social sciences because it translated a previously abstract notion scattered across disciplines that was lacking a common conceptual foundation with an operationalizable analytical pipeline. With his structured research paradigm, Entman introduced a framework that can be applied to measure the same concept across fields by identifying frames through observable textual patterns such as keywords or particular sentence construction techniques. This framework was what initially enabled human annotation to achieve meaningful intercoder agreement and is what opens the door to exploring computational mechanisms for detection at scale.

Entman's approach is particularly valuable in the context of political news, where framing has been shown to shape the major manifestations of public opinion itself (Entman, 1993). As he argues, framing in that context carries important implications for how political power is exerted. Examining which elements receive attention and which are directed away from it can help illuminate the power dynamics embedded in editorial choices.

Entman emphasises that what is salient to an audience is a product of the interaction between text and receivers. Since the presence of a frame in a text does not guarantee its influence on audience's thinking, the scope of this study aims to analyze framing intent from the communicator side rather than as a measured effect on readers.

2.2. Media Framing and the Conflict Frame

The operationalizable approach to framing proposed by Entman inspired subsequent efforts to identify specific recurring frames. This occurred particularly in news coverage where, as established above, its implications for public opinion and political power are significant. Building on earlier work (Iyengar, 1991; Neuman et al., 1992), Semetko & Valkenburg (2000) made a significant contribution by proposing a standardised deductive set of indicators to measure their prevalence. This approach allows it to be replicated easily, cope with large samples and detect differences in framing within media outlets. Beyond its theoretical value, the operationalization of frames as discrete binary questions proved particularly useful to define a clear with a few-shot annotation prompt.

Among the identified frames, the conflict frame stands out for its prevalence and consistency across contexts. Defined as coverage that emphasizes conflict between individuals, groups, or institutions as a means of capturing audience interest, it has been documented as one of the most prevalent frames in political news (Semetko & Valkenburg, 2000). Its relevance lies in the fact that exposure to critical, conflict-oriented news coverage has been shown to reduce trust in political leaders (Kleinnijenhuis, van Hoof, & Oegema, 2006).

The conflict frame belongs to a wider research tradition shared with other prominent frames, often labeled interchangeably as strategic or game framing (Cappella & Jamieson, 1997; Aalberg et al., 2012; Strömbäck, 2008). This body of work examines political journalism organized around the tactical maneuvering of actors such as who is winning, who is losing, who is attacking whom, rather than policy substance. The conflict frame used in this study is analytically narrower than this tradition, but shares its core structural requirement: identifiable actors in opposition must be present. This means that a coverage centered on political personalities and institutions provides more room for the conflict frame and the broader strategic/game tradition to appear than coverage centered on technical or procedural content.

The selection of the conflict frame for this study was not arbitrary. It reflects a hypothesis about the Chilean media ecosystem, one that the empirical evidence reviewed in the following section supports.

2.3. Conflict Framing in Chilean Political News

In a country where newspapers have successfully operated their digital transition, reaching millions of unique monthly visitors (Reuters Institute Digital News Report, 2026), the Chilean press has been extensively studied. While initial studies focused on ownership concentration and editorial homogeneity (Becerra & Mastrini, 2019; Mellado & Rafter, 2014), research has later examined if this homogeneity also extended to how the press frames political news.

Gronemeyer & Porath (2017a) found that conflict was the most prevalent frame (mean intensity score of 0.49) in a corpus of 1,164 news items from six Chilean newspapers across three sampled years (2007, 2011, and 2015). Following Semetko and Valkenburg's (2000) approach, this score is built by averaging the activation of each frame's question across all sampled articles, ranging from 0 (frame entirely absent) to 1 (all indicators present). The relatively high standard deviation ($SD = 0.37$) in comparison to the mean indicates substantial variation across articles.

Chile also presents an interesting case for media studies due to its documented gap between citizens' perceptions and empirical reality across multiple social issues. On crime, the Government's own National Urban Survey on Citizen Security (ENUSC) found that 87.6% of Chileans perceived rising crime in 2023, even as actual victimization fell to a historic low of 21.7%. This gap is partially attributed to media amplification of high-impact crime coverage (CEP, 2024). This disconnect has also been studied around the topic of migration. Ajzenman et al. (2023) show that a rise in immigration had no effect on actual crime rates yet significantly increased crime-related concerns, while Scherman et al. (2022) demonstrate experimentally that exposure to negatively framed migration news directly causes more intolerant attitudes toward migrants.

Another distinct characteristic of Chile for conflict framing studies is the number of high-intensity political events the country has experienced over the past decade. The 2019 *estallido social*, the two constitutional plebiscites, the 2021 and 2025 presidential elections, among others, were key political events and excellent case studies to examine whether media conflict framing is associated with moments of heightened political relevance. In this line, Matus, Vitores, and Ramírez Friderichsen (2024) found that conflict was the dominant frame in Chilean front-page coverage of the *estallido*, ahead of human interest and attribution of

responsibility. Further comparisons with other political periods, conducted at scale, may allow researchers to test whether this is a constant or reliable feature of Chilean political coverage or one that is specifically associated with this politically charged period.

2.4. LLMs as at-scale Annotators

Historically, large-scale framing analysis has depended on having access to a manually annotated corpus, such as the Media Frames Corpus (Card et al., 2015). This requirement has been a significant barrier for languages other than English, where no such resource exists and building one requires annotators with sufficient time and knowledge (Pangakis & Wolken, 2024). Gilardi, Alizadeh, and Kubli (2023) found that LLMs offer a way around this barrier. Across several annotation tasks, including frame detection, ChatGPT outperformed crowd-workers and matched the intercoder agreement of trained human annotators in a zero-shot setting. Alizadeh et al. (2023) extended this comparison to open-source models using both zero-shot and few-shot approaches, finding that few-shot prompting further improved performance. Both studies opened the possibility of testing whether these results extend to Spanish-language news and to the conflict frame specifically, neither of which has been examined in this literature.

Even though LLM-based annotation already allows researchers to work at scale, running a large model over hundreds of thousands of documents remains computationally expensive and time-consuming. The open-source model approach offers advantages in cost, transparency, and reproducibility over proprietary alternatives. For this reason, this study relies on a QWEN model for its annotation pipeline. Another approach to reduce costs and increase accessibility is the one followed by Pangakis and Wolken (2024), who address this by using the LLM-generated labels to fine-tune a smaller classifier that approximates the larger model's performance. This raises the question of whether this additional step should also be considered when building an annotation pipeline, or whether there is a performance loss worth considering.

3. Methodology

3.1. Corpus presentation

The initial corpus, *prensa_datos* (Olea, 2026), was provided by Bastián Olea and corresponds to his own database of Chilean news media built using systematic scraping. It contains 899,545 articles from 27 outlets, from 2007 to April 2026.

For the purpose of this research, several modifications were made to the original corpus. First, the dates were filtered from 2018 to 2026, to reduce the scope and increase granularity of the analysis.

BioBioChile was added to the corpus given its national relevance. It ranks among the most trusted outlets in Chile (65% trust score, Reuters Institute Digital News Report 2025) and is the highest-traffic Chilean news site, with approximately 26 million monthly visits (Similarweb, 2026).

Ten outlets were excluded for the following reasons: insufficient temporal coverage for time analysis (24horas, La Hora, El Siglo, Exante), scope outside political-informational journalism (Redgol), specialized formats such as investigative journalism (Ciper) and opinion columns (Quinto Poder) and insufficient volume for reliable statistical representation after temporal filtering (CHV Noticias, La Nación, La Cuarta).

Since the corpus contains articles covering all news domains, it was further filtered to retain only politically-oriented articles. While conflict framing can appear across all news domains, in political journalism it reflects editorial choices about how political tensions are represented, which is the core object of this study. Retaining non-political content when analysing this frame at scale would introduce noise without adding interpretive value.

3.2. Analytical Pipeline

To ensure full replicability of the analytical pipeline for future research, all methodological decisions prioritized open-source and computationally accessible techniques.

3.2.1. Corpus Filtering via Thematic Clustering

The political subcorpus was selected through unsupervised thematic clustering. TF-IDF was preferred over dense embedding-based representation for three reasons: interpretability (clusters are labeled by their highest-weight terms), computational feasibility (operations can be performed without GPU resources), and suitability for distance-based clustering (they avoid the curse of dimensionality, which tends to collapse cluster boundaries; Beyer et al., 1999; Radovanović et al., 2010).

Each article was represented as a TF-IDF vector (Salton & Buckley, 1988) concatenating the title and body text. Dimensionality reduction was then applied via Latent Semantic Analysis (LSA; Deerwester et al., 1990) using the TruncatedSVD implementation from the scikit-learn library (Pedregosa et al., 2011), projecting the vectors into a dense semantic space of 500 dimensions.

The dense LSA vectors were clustered using MiniBatch K-Means (Sculley, 2010). It was selected over standard K-Means for computational feasibility due to the size of the corpus and its availability in the scikit-learn library. The normalized vectors produced by LSA make Euclidean distance approximate cosine similarity, which makes K-Means geometrically appropriate for this representation. The specific parameters used at each stage of the corpus filtering pipeline can be found in Appendix 1.

To decide the cluster size, a K grid search was performed over a $K \in \{20, 30, 40, 50, 60, 70, 80\}$. The quality of the clustering was evaluated using the Silhouette Score (Rousseeuw, 1987) with cosine metric over a stratified sample. $K = 70$ was selected as the marginal improvement in Silhouette Score beyond this value was below the defined threshold ($\Delta < 0.001$).

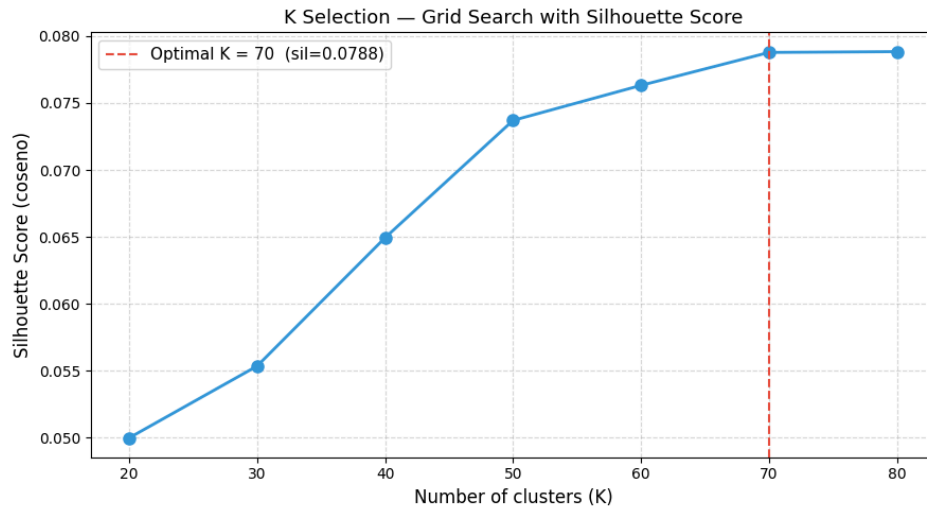


Figure 1. Silhouette score by number of clusters (K), with optimal K selection

Each cluster was manually labeled by inspecting the 15 highest-weight terms in the centroid. In cases where the terms were not sufficiently interpretable, an LLM-assisted review of the content of the articles was performed. Final labeling decisions were made by the researcher.

To decide whether the cluster was going to be considered to analyze the conflict frame, the Master Codebook of the Comparative Agendas Project (Jones et al., 2025) was used as a framework. This codebook provides 21 thematic categories for policy process studies and has been used in diverse cases of manual codification of political documents, including media content.

After this categorization, the subcorpus includes 734,500 articles, considering all clusters whose dominant theme corresponds to one CAP category. This decision was made under the assumption that these themes constitute the domain where political dispute, and therefore conflict framing, is most likely to be observed in journalistic coverage. The final clusters with their corresponding label, top words and CAP topic can be found in Appendix 2.

An initial definition of political content, limited to a small set of manually selected explicitly political topics, was also tested. Its narrowness made it both less replicable and less representative of the full range of political journalism, which directly affected the proportion of articles identified as containing the conflict frame. This underscores how sensitive this metric is to how the corpus is defined.

3.2.2. Synthetic Annotation using an LLM

The entire political subcorpus (734,500 articles) was annotated using a QWEN-based model with few-shot learning. This pipeline avoids depending on proprietary APIs and allows it to be fully reproduced without recurring inference costs, which is a relevant consideration given the corpus size. The following sub-sections describe how the conflict frame was operationalized and translated into annotation instructions for the model.

3.2.3. Model Implementation and Inference Parameters

The annotation pipeline was implemented using QWEN3-8B, served locally through vLLM (Kwon et al., 2023), an inference engine optimized for high-throughput LLM using techniques such as continuous batching. This was a practical requirement given the corpus size, because sequential inference would have made annotating 734,500 articles computationally infeasible within the resources available for this study. This setup allowed it to run the inference process on the full corpus in a single A100 GPU in approximately 30 hours.

Two model-level decisions are worth specifying. First, generation was run at temperature 0.0 to maximize determinism and reduce run-to-run variability in the JSON outputs. Second, QWEN's "thinking" (extended reasoning) mode was disabled during inference. This choice was made primarily for processing speed, given the scale of the task and the time constraints of the project. It is relevant to consider that enabling extended reasoning could potentially improve performance and annotator agreement on the more demanding tasks where the model struggled, such as Q3. The difficulties with this dimension are discussed in the following section. The trade-off between the potential gains in accuracy and a higher inference time remains as an open question for future validation.

Another important decision was the length of text provided to the model as input. Each article was represented to the model as its headline plus a lead extract of up to 1,800 characters from the body text, truncated at the nearest sentence boundary. This truncation was a deliberate design choice justified mostly by the fact that using full articles caused GPU memory errors when running 32 concurrent workers, making it a requirement to sustain stable inference across the full corpus. While this was primarily a technical constraint, its impact could be mitigated when considering the assumption that news writing is conventionally taught to present the most newsworthy information at the beginning of the

article. This would mean that the lead extract captures most articles' core content even without the full text. Still, this premise is not guaranteed to hold uniformly across all articles and outlets in the corpus. For this reason, it is important to consider that this approach may lose relevant context in longer articles, where relevant conflict appears later in the text.

3.2.4. Conflict Frame Operationalization

Following Semetko & Valkenburg (2000), the conflict frame was operationalized based on their deductive approach, using as a base the following three questions:

- Q1: Does the story reflect disagreement between parties-individuals-groups-countries?
- Q2: Does one party-individual-group-country reproach another?
- Q3: Does the story refer to two sides or to more than two sides of the problem or issue?

The fourth question proposed by the paper (Does the story refer to winners and losers?) was discarded, as the validation study found it did not correlate sufficiently with the other conflict indicators.

Translating the three remaining questions into annotation criteria required an iterative process, as the three items posed different challenges for the model. Q1 and Q2 proved equally straightforward, requiring only one significant refinement. A centrality rule was added, specifying that disagreement or reproach must be the narrative core of the article, not incidental context used to explain another event. This rule was necessary because, unlike a human annotator who would naturally discount a passing mention, the model tended to activate the dimension even when disagreement or reproach appeared only as a minor detail unrelated to the article's main focus. Both items were also clarified as not requiring a direct quotation, to avoid the model overcorrecting toward only recognizing disagreement or reproach when explicitly quoted.

Q3 was considerably more difficult. While identifying two sides of an issue might be a relatively simple task for a human annotator, the model struggled to recognize what actually constituted opposing sides. Two recurring types of false positives were identified: (a) articles featuring two actors with aligned perspectives on the same topic, and (b) paraphrased opinions attributed to an actor as if they were direct quotes. To address this, the

operationalization of Q3 evolved through several iterations. An initial requirement that both sides have an attributed position, whether through direct quotation, indirect statement, or explicit description, proved insufficient. Under these conditions, the model still activated Q3 simply because one actor mentioned or reproached another, even when that other actor's own position was never presented in the text. This led to a stricter rule: the model was required to identify a quotable statement from each of two distinct actors before activating Q3. The final version reformulated this as a two-step verification protocol: first confirming the presence of one actor's own quoted position, then confirming a second, distinct actor's quoted position that opposes the first on the same topic. This protocol was accompanied by an explicit opposition test (does the second position genuinely tension the first, or merely coexist with it?). A list of exclusions was also included, and the model was instructed to disregard two actors from the same side, someone expressing an opinion over an institutional actor or any actor mentioned in the article without expressing a political position of their own. Even with these adjustments, the model was not able to perform this task reliably across all cases, resulting in markedly low agreement levels with human judgment (reported in the Results section) and hallucinations or inconsistencies regarding the actors and quotes that the model identified. This limitation likely reflects two factors: the model's lack of background knowledge about which political actors are opponents and the difficulty of tracking full semantic context across longer truncated text.

Beyond the rules applied to each question, the construction of the few-shot examples themselves proved to be a critical component of the prompt. The final prompt included 14 examples, each curated to represent complex cases identified by the researcher during the iterative review process as instances of recurrent misjudgment:

- Ceremonial statements and verifiable institutional facts that should not activate any dimension
- A single actor narrating or filtering another's position, rather than that position being independently expressed
- Legislative or administrative procedures reported as routine fact despite an underlying political disagreement
- Interactions between two real actors that resolved in agreement rather than confrontation

- Disagreement between actors from different countries, included to broaden the diversity of scenarios represented

Finally, output format mattered just as much. The model was instructed to return a fixed JSON object for each dimension, which proved effective for systematizing the information for later analysis. Outputs that could not be parsed as valid JSON were flagged and excluded from the final frequency calculations. Only 0,98% of articles in the 734,500-article batch fell into this category, leaving a final corpus size of 727,113 articles.

Ultimately, this entire process reflects a human-in-the-loop, iterative approach to prompt design, similar to validation procedures documented in other LLM-annotation pipelines (Pangakis & Wolken, 2024), though in this case the tuned LLM served as the final annotator rather than a source of training labels for a smaller fine-tuned classifier. Identifying which cases the model was likely to misjudge, curating examples that addressed them and refining the prompt's rules accordingly prove to require deep knowledge of the corpus.

3.3. Encoder Fine-Tuning: A Discarded Approach

As a complement to the LLM-based annotation pipeline, the feasibility of a lighter classification model, XLM-RoBERTa-base, was explored. The encoder was fine-tuned using 10,000 articles previously labeled by QWEN (silver labels). A separate set of 146 manually produced gold annotations was used as a validation to assess how well the fine-tuned model's predictions matched human judgment. The annotation results, contrasted with the human sample, led to reconsidering including an encoder into the pipeline. The experiment empirically confirmed the warning raised by Lu & Smith (2025) in "Feeding LLM Annotations to BERT Classifiers at Your Own Risk" about the structural risks of error propagation when training classifiers on LLM-generated annotations without sufficient human correction. The three dimensions were overestimated in most cases. Bucher & Martini (2024) show that small fine-tuned models trained on human gold labels do consistently outperform LLMs used in zero/few-shot settings, indicating that the problem was not fine-tuning itself, but the training with silver labels due to the insufficiency of the gold set (146 versus the ~500–1,000 recommended).

Given the volume of manual annotation that would have been required to specifically resolve dimension Q3 and the time available for this investigation, the decision was made to discard

the fine-tuned encoder as the final classifier and adopt few-shot LLM prompting directly on the corpus instead. Further research could explore whether fine-tuning on a larger gold set is able to resolve the Q3 calibration or improve annotator agreement on Q1 and Q2 beyond their current borderline levels. If so, a gold-calibrated fine-tuned encoder could offer a faster and less resource-intensive alternative to annotating the full corpus with the larger LLM.

4. Results

This section presents the study's findings in two parts. First, it evaluates the annotation pipeline's reliability against human-coded ground truth. This allows to establish the extent to which the model's judgments can be trusted before any substantive claims are drawn from them. Second, building on this validation, it presents a large-scale analysis of conflict framing across the Chilean political news corpus (2018–2026) as a case study.

4.1. Model Performance

To assess whether LLMs can reliably identify conflict frames in Spanish-language news articles at scale, the annotation pipeline's performance was validated against a human annotator. A sample of 296 articles, stratified by the presence of the conflict frame and overrepresenting the minority class (conflict frame present), was coded using the same prompt as a rubric.

Dimension	Raw Agreement	Human Rate	Model rate
Q1	80.1%	31.8%	41.6%
Q2	84.1%	33.4%	29.1%
Q3	82.1%	9.1%	19.6%

Table 1. Raw agreement between model and human annotator by conflict frame dimension

Note. Q1 = disagreement between parties; Q2 = reproach; Q3 = two sides of the issue.

The pipeline's raw agreement with the human annotator, was benchmarked against the most comparable prior human-annotated conflict frame study in Chile. Gronemeyer and Porath (2017a) report human intercoder reliability of 0.757 for Q1, 0.891 for Q2, and 0.771 for Q3, using Holsti's coefficient, equivalent to raw agreement in this case. By this measure, the

pipeline's agreement with the human annotator is comparable to the agreement reached between two human coders in their study across three dimensions.

It is worth noticing that raw agreement doesn't separate real agreement from agreement that would happen just by guessing "no" most of the time. This is relevant in this case because, as shown in following sections, most articles have no conflict frame at all. For this reason, Krippendorff's alpha was also calculated. This metric corrects this phenomenon by discounting the agreement expected by chance, making it a more conservative and appropriate measure of substantive agreement. Following conventional thresholds for content analysis using this metric (Krippendorff, 2004), values above 0.80 are considered sufficient for confident conclusions, values between 0.67 and 0.80 are acceptable for tentative conclusions, and values below 0.67 are considered insufficient.

Dimension	Krippendorff's α
Q1 (disagreement)	0.572
Q2 (reproach/criticism)	0.631
Q3 (two sides)	0.273

Table 2. Krippendorff's alpha by conflict frame dimension

These results indicate that Q1 and Q2 ($\alpha = 0.572$ and 0.631 , respectively), the two dimensions identified in Section 3.2.4 as more straightforward for the model to operationalize, perform better than Q3, but still fall short of the 0.667 threshold for tentative conclusions when considered individually. Q3 ($\alpha = 0.273$), by contrast, falls well below either dimension, confirming the difficulties described earlier regarding the model's ability to verify opposing positions. Given this low reliability, Q3 was excluded from the composite indicators used in subsequent analyses.

It is worth noting that conflict framing was a challenging coding task even for human annotators. In the course of producing the gold annotations for this study, the most persistent ambiguity was to identify if the tension present in an article was directed to another actor or it was more a criticism or institutional friction without a clearly defined target. The ideal validation for this pipeline would involve comparing the model's annotations against a second

independent human coder, to minimize the bias of a single annotator's judgment. Using human-human intercoder reliability would provide a more appropriate benchmark than treating one person's labels as ground truth. This was not feasible within the scope of this study.

4.2. Analytical Metrics construction

As Q1 and Q2 did not individually reach sufficient reliability, and considering that the theoretical framework treats the sub-dimensions as correlated indicators of a single underlying construct (Semetko & Valkenburg, 2000), composite measures were tested as an alternative operationalization.

The intensity score follows the implementation of Semetko and Valkenburg (2000). It is calculated by averaging the binary scores of the questions that constitute that frame, producing a continuous range from 0 (frame absent) to 1 (frame fully present). The binary indicator, by contrast, defines if an article contains a conflict frame or not by analyzing if either Q1 or Q2 was activated. These two metrics are complementary. The binary indicator captures how widespread the conflict frame is across the corpus, while the intensity score reflects how strongly the frame is expressed on average, since articles with both dimensions contribute more to the mean than articles with only one.

Dimension	Krippendorff's α	Raw agreement
Binary	0.696	85,1%
Intensity Score	0.714	70.6%

Table 3. Krippendorff's alpha and raw agreement for composite conflict frame indicators

The binary composite indicator reached $\alpha = 0.696$, and the intensity score $\alpha = 0.714$, both exceeding the threshold for tentative conclusions, despite neither individual sub-dimension doing so on its own. This indicates that human and model annotators converge more reliably on the overall presence and degree of conflict than on which specific sub-dimension is driving that judgment. Given these results, both metrics were retained for the analyses that follow. The binary indicator is used as the primary descriptive measure given its more direct interpretability as a prevalence rate, while the intensity score is retained specifically for

comparison with prior literature using the same scale (Semetko & Valkenburg, 2000; Gronemeyer & Porath, 2017a) and for analyses of dispersion across outlets or clusters.

4.3. Conflict Framing in Chilean Political News, 2018–2026

Having established that the annotation pipeline performs adequately enough on a binary setting, this section presents a large-scale analysis of conflict framing in the Chilean political news corpus.

4.3.1. Corpus-wide Prevalence

Across the full annotated corpus, the binary indicator showed a conflict frame prevalence of 28.3%, implying that more than one in four politically-themed articles contains some form of conflict framing. The intensity score offers a complementary view of the same pattern, with a mean of 0.24 and a standard deviation of 0.40. This also shows consistency with the binary indicator, given that 71.7% of articles have no conflict-frame activation at all. Taken together, these metrics show that the conflict frame is concentrated in a specific, relatively small segment of the corpus.

By comparing this result with prior estimates for Chilean political news, it produces notably different results. Gronemeyer and Porath (2017a) reported a conflict frame intensity mean of 0.49, roughly twice the value found in this study. The most likely driver of this difference is how the two studies define the political subcorpus over which prevalence is calculated. This study uses an unsupervised, CAP-based political subcorpus that considers a broad range of domains (e.g., Health, Education, Macroeconomics). On the other hand, Gronemeyer and Porath (2017a, 2017b) defined their political subcorpus narrowly, limited to news items addressing government affairs, the political elite, political parties or social movements pressuring for policy change. This difference likely affects the proportion of articles identified as containing the conflict frame. This hypothesis is examined further below, where conflict frame prevalence is broken down by thematic cluster.

Two other factors are worth considering. First, this study's measure averages only two items (Q1, Q2), while Gronemeyer and Porath's averages three (Q1, Q2 and Q3). Given Q3's low reliability, this study cannot meaningfully estimate how its inclusion would have affected the score. Second, the studies address different time periods where conflict framing prevalence may itself have shifted. This latest hypothesis cannot be evaluated with the data available.

4.3.2. Variation by Topic

Breaking down conflict framing intensity by topic shows substantial variation across categories.

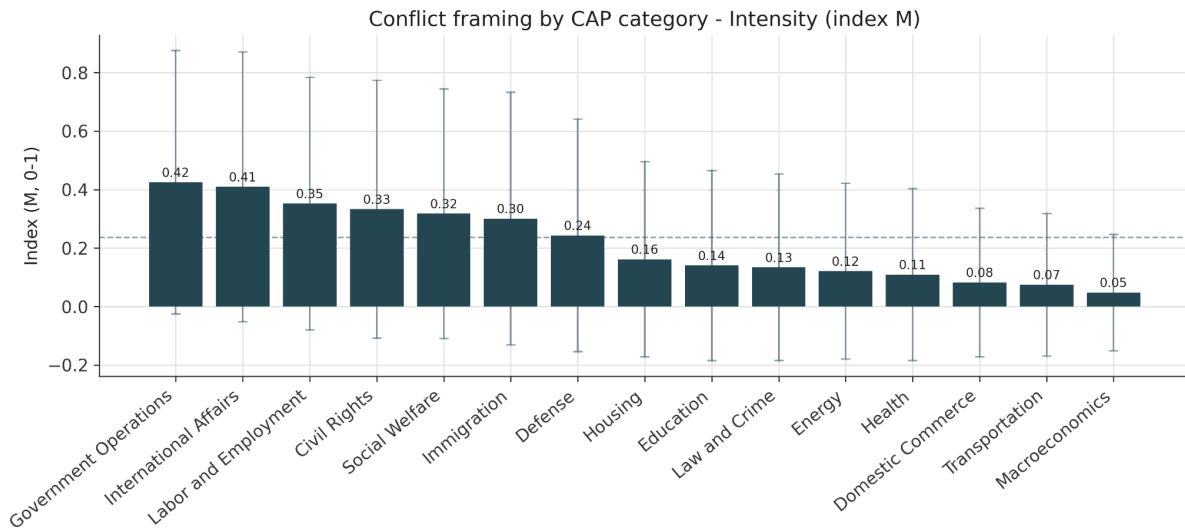


Figure 2. Conflict frame intensity by CAP thematic category

Government Operations, a category closer to Gronemeyer and Porath's (2017a) narrower corpus definition based on government affairs and related topics, shows an intensity of 0.42. This value is much closer to their reported value (0.49) than the corpus-wide mean. Macroeconomics, by contrast, has an intensity of just 0.05. This shows that the corpus-wide average of 0.24 masks just how unevenly the conflict frame is distributed across topics, reinforcing how much a study's reported intensity depends on which categories its subcorpus includes.

Within *politica_parlamentaria_interna* (part of Government Operations category), articles classified with conflict frame are marked by explicitly confrontational vocabulary (*crítico, reprochó, ultimátum*), while index=0 articles cluster around routine legislative administration (*viáticos, inasistencias, circunscripción*). The same phenomenon appears within the low-intensity *pandemia_covid_sanitaria_cluster* (Health). The minority of index=1 articles are precisely those where pandemic coverage turned into political dispute (*acusó, oposición, Jadue, UDI*), while index=0 articles are dominated by unrelated sanitary reporting (avian flu outbreaks, shellfish contamination, hantavirus). Conflict activation, in other words, tracks whether an article's content is itself adversarial, rather than its cluster's assigned topic. This

supports the interpretation that category-level differences in conflict framing are substantive rather than an artifact of how the political subcorpus was constructed.

Clusters were manually classified as actor-driven if their dominant vocabulary centers on human actors, individual or collective, as protagonists of the narrative, as opposed to clusters organized around the substance of a policy. Under this classification, 13 of the 55 clusters (215,221 articles, 29.6% of the corpus) are centred on an actor's behaviour, while the remaining 42 (511,892 articles, 70.4%) are theme- or process-oriented.

Crime and immigration are worth examining specifically because both have been linked to a documented gap between media coverage and underlying reality. In the case of crime, the Law and Crime category is the largest one in the corpus by volume (206,708 articles) and also one of the lowest in aggregate activation (16.2%). Disaggregating by cluster reveals that activation concentrates almost entirely in clusters involving institutional or elite actors: *casos_judiciales_alto_perfil* (40.1%), *seguridad_politica_interior* (39.9%), and *poder_judicial_corte_suprema* (35.4%). On the other hand, clusters covering routine crimes such as robberies, homicides or drug trafficking, barely register on the conflict frame (*robos_delincuencia*: 0.7%; *homicidios_pdi*: 0.7%; *narcotrafico_drogas*: 1.5%).

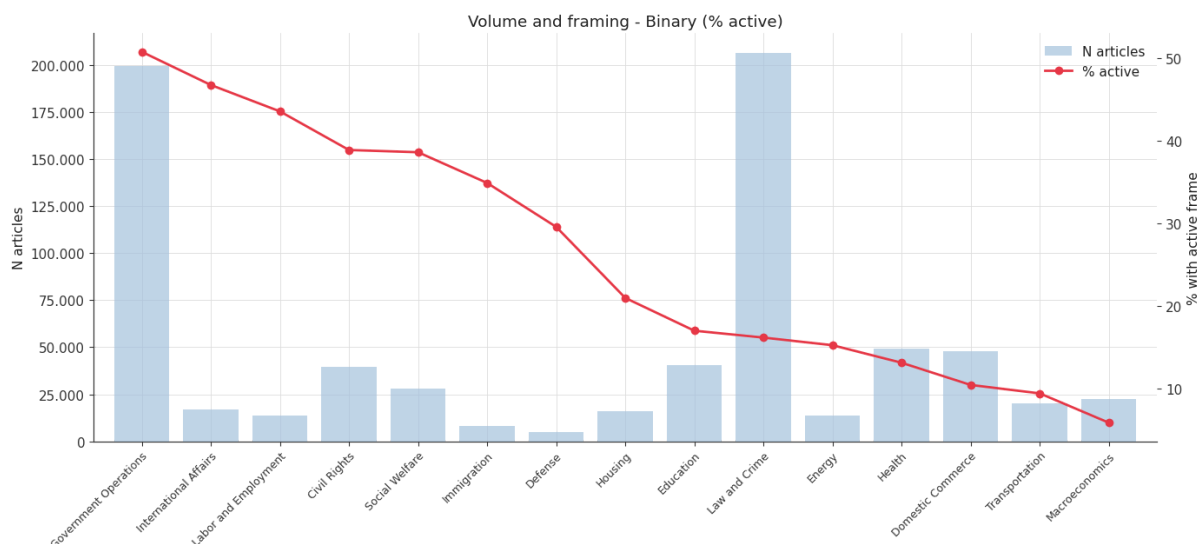


Figure 3. Article volume and conflict frame prevalence by CAP category

Immigration shows the same underlying division as crime: articles classified as containing conflict frame are marked by named legislators clashing over migration policy (*Kaiser*,

Schalper, Longton, oficialismo, bancada, crítico), while articles without it are dominated by administrative and border-control reporting (detentions, nationalities, police operations). The difference from crime, then, is not that migration coverage is inherently more politicized as a topic, but that a substantially larger share of migration coverage falls on the dispute side.

4.3.3. Variation by Outlet

Having established that conflict activation tracks actors rather than topics, a further question is whether outlets apply the frame uniformly. At first glance, the difference is substantial. Across the 18 outlets in the corpus, the binary indicator ranges from 15.4% (MegaNoticias) to 45% (IzquierdaDiario).

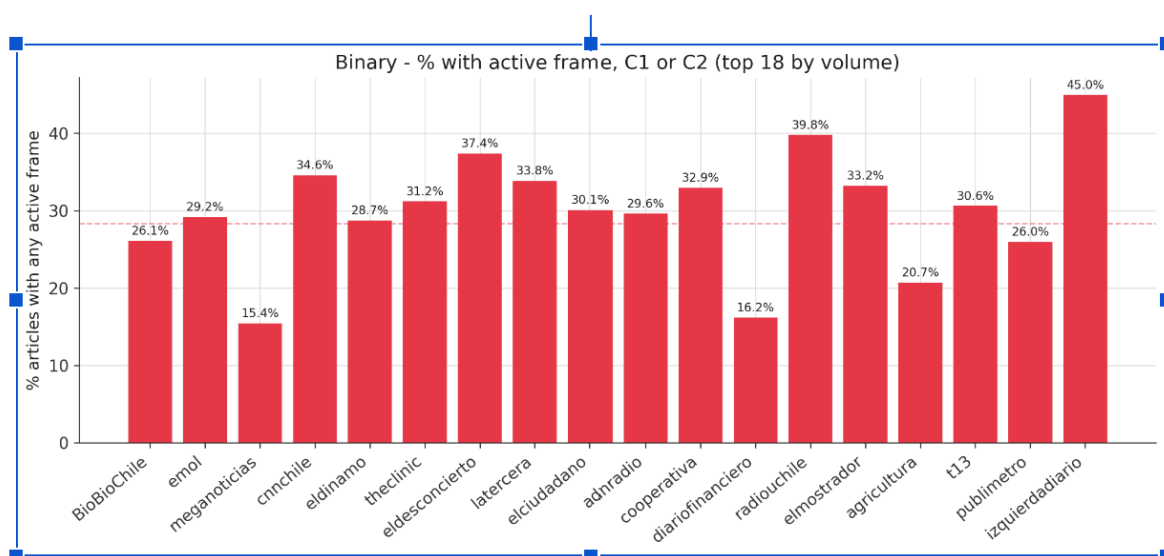


Figure 4. Conflict frame prevalence by outlet

Nevertheless, on its own, this raw comparison says relatively little about outlets' framing behavior, since they differ substantially in what they cover. *Diario Financiero*'s coverage, for example, is based mainly on Domestic Commerce and Macroeconomics. These are precisely the categories shown above to rarely activate conflict framing, regardless of outlet. Whether *Diario Financiero* covers its few contested topics any less confrontationally than other outlets is a separate question this aggregate comparison cannot answer. It requires controlling for what is being covered, which is why a category and cluster level analysis is required.

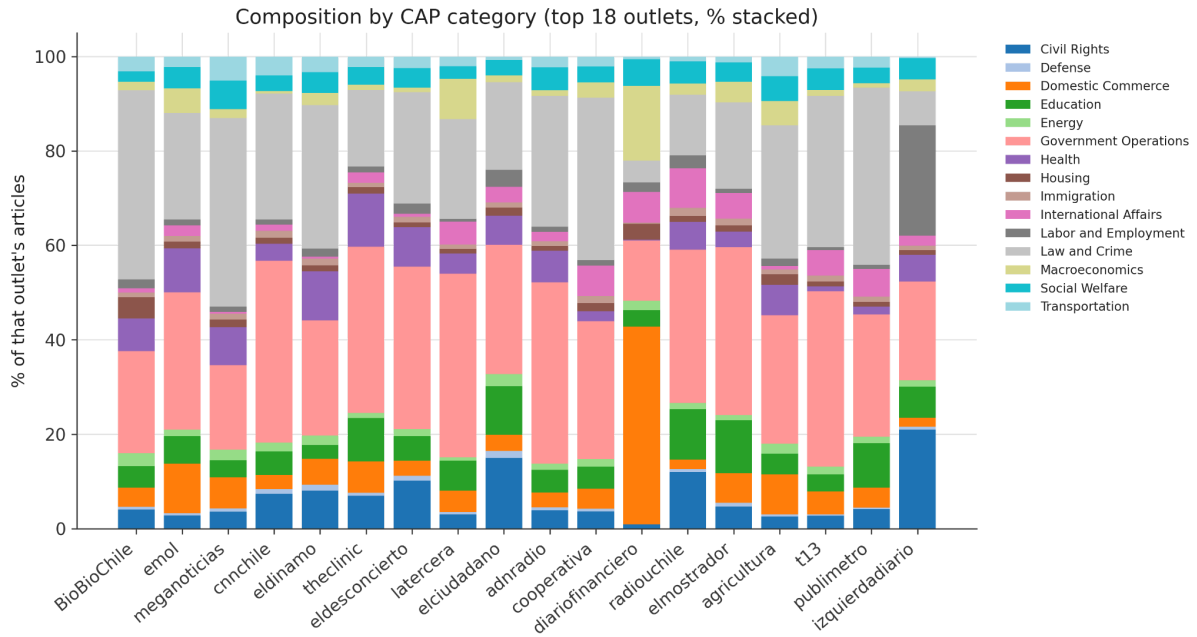


Figure 5. Thematic composition of coverage by outlet

The comparison at the cluster level reveals that editorial divergence between outlets is not random across topics. Clusters where outlets diverge most are, almost without exception, the same clusters with the highest overall activation ($r = 0.65$ between a cluster's mean intensity and the standard deviation across outlets). This indicates that outlets broadly agree on which topics are conflictive, but vary considerably in how intensely they apply that shared judgment.

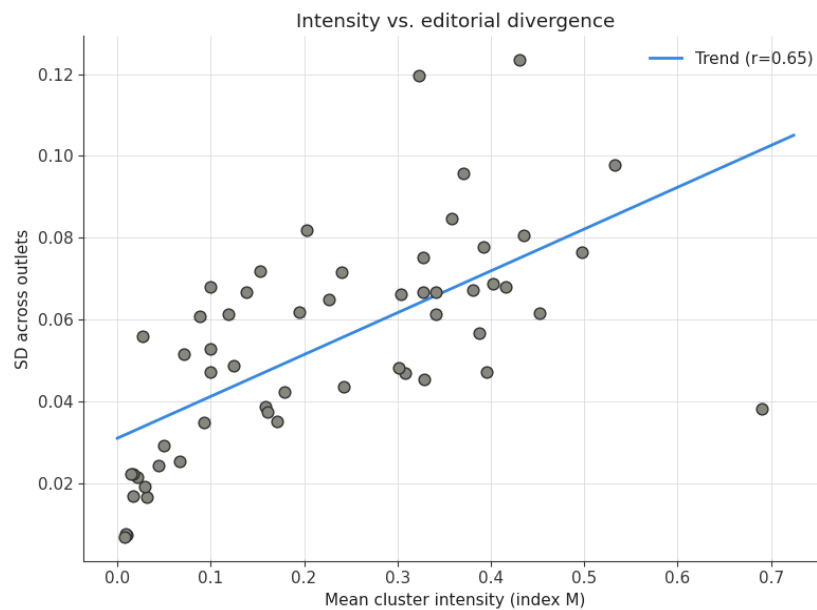


Figure 6. Cluster-level conflict intensity versus cross-outlet divergence

Retiros_fondos_afp, the cluster with the highest between-outlet divergence in the corpus, illustrates this rule directly. Coverage of the pension fund withdrawal debate ranges from an index of 0.19 (Agricultura, n=220) to 0.55 (El Mostrador, n=143), nearly a three-fold difference in how confrontationally the same policy issue is covered.

A notable exception to this rule is *politica_parlamentaria_interna*, which has the highest mean activation of any cluster in the corpus (0.69) yet one of the clusters with lower levels of divergence. This means that outlets across the ideological spectrum cover routine parliamentary confrontation as conflictual.

4.3.4. Variation by time period

While the patterns above describe what drives conflict framing, a separate question is when it concentrates. Section 2.3 left open whether the dominance of the conflict frame found by Matus, Vitores and Ramírez Friderichsen (2024) during the *estallido social* is a constant feature of Chilean political coverage or if it varies in politically charged period. This section addresses that question directly by examining whether the frame's prevalence intensifies around other moments of heightened contestation, such as elections or social protest.

To assess whether the binary conflict frame indicator shifts around political events beyond ordinary month-to-month variation, each event's pre- and post-event window (1, 3, and 6 months) is compared against the full distribution of equivalent-length windows across the entire series. This yields a percentile for each event. A shift at the 95th percentile, for instance, is larger than 95% of all comparable windows observed anywhere in the corpus, providing a more conservative benchmark than comparing against a single summary statistic such as the median.

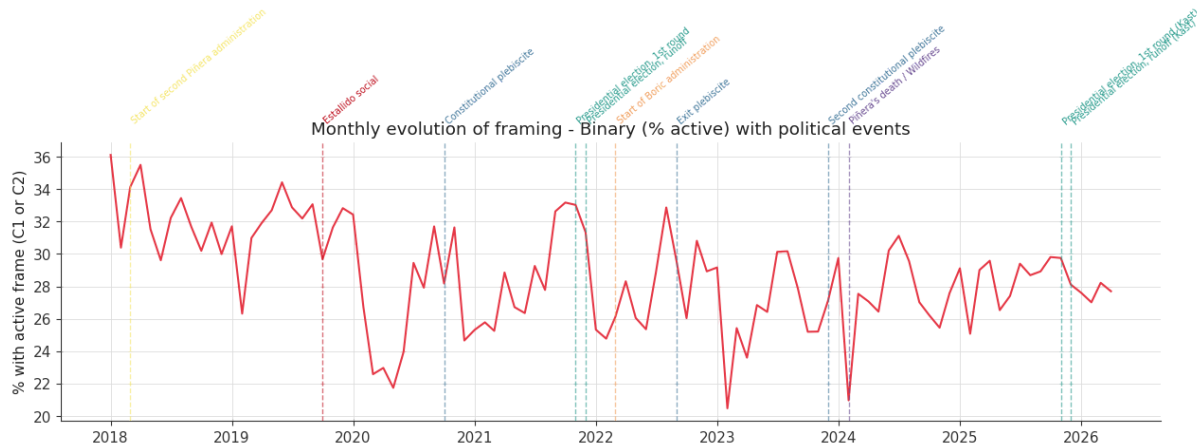


Figure 7. Monthly evolution of conflict frame prevalence with political events overlaid

Applying this approach across the eleven events under study (full results in Appendix 3) shows that conflict frame activation does not track major political events uniformly, though the pattern is more nuanced than a simple presence or absence of elevation.

The *estallido social* stands out as the most consistent episode in the entire series. Activation is elevated across pre-event windows of 1, 3, and 6 months simultaneously (91.8th–96.8th percentile of all comparable windows in the series). It is worth noting that part of this may reflect 2018–2019's systematically higher baseline rather than the *estallido social* itself. Given the uprising's unforeseen nature, this elevation cannot be read as anticipation of the specific event.

Among electoral events, the 2021 presidential election is the clearest case of pre-election elevation, present in both the first round (69th–92nd percentile) and runoff (76th–93rd percentile). The 2025 election cycle, by contrast, shows no comparable elevation in any window (56th–67th percentile, consistent with normal month-to-month variation).

Government transitions show no single pattern. While the start of the Boric administration in 2022 shows no comparable rise in any window (9.1th–73.4th percentile, within normal variation), the start of the second Piñera administration in 2018 cannot be meaningfully assessed with this method, because part of the pre-transition windows fall outside the corpus scope. Its high percentile in the month following the transition (98.0th) likely reflects 2018–2019's elevated baseline rather than a transition-specific spike.

Beyond discrete political events, the series shows a consistent seasonal pattern: February is the lowest-activation month of the year in every year of the corpus (mean = 0.210, compared to 0.246 in January and 0.233 in March), with low year-to-year variation (SD = 0.027) indicating this is a recurring calendar effect rather than a coincidence tied to any single year. This calendar effect has a mirror image at its opposite extreme: July and August are the two highest-activation months of the year (mean = 0.301 and 0.300, respectively), with July also showing the lowest year-to-year variation of any month (SD = 0.015), the most consistent month in the entire series.

5. Discussion and conclusions

This study was designed to answer two questions: whether an human-in-the-loop LLM annotation pipeline can reliably detect conflict framing in Spanish-language news at scale, and what its application reveals about this frame in Chilean political coverage.

On the first question, the pipeline was found to achieve a level of agreement sufficient to support the descriptive analysis that follows, even though there is clear room to improve it further. None of the three conflict-frame dimensions met this standard on its own, and Q3 in particular, was excluded from the substantive analysis. Reliability only reached an acceptable threshold once Q1 and Q2 were aggregated into the binary and intensity indicators, meaning the model is better at detecting the overall presence of conflict framing than at attributing it to a specific mechanism (disagreement versus reproach). This qualified level of reliability, combined with the fact that it reflects agreement with a single annotator rather than a consensus between independent coders, justifies treating the pipeline's output as adequate for the corpus-level analyses that follow while cautioning against over-interpreting any single article-level classification or fine-grained sub-dimensional pattern.

Beyond the evaluation of the model's performance, the pattern across the conflict frame dimensions corroborates, in a new language and text format, what prior work on LLM annotation has suggested. Dimensions signaled by specific verbs and tone within the sentence itself, as is the case with disagreement and reproach in the conflict frame, tend to be more reliably annotated than dimensions requiring a higher level of relational inference. Determining whether two positions are genuinely opposed is rarely a literal reading of the text and it often requires knowing who is speaking and where they stand politically,

background knowledge the model may not have. Tasks of this second kind remain unreliable at least under the conditions under which this study was conducted.

Finally, the contribution of this study, more than the LLM annotator, is a full pipeline for conflict frame analysis at scale. Several components of this process are directly reusable as specified, such as the CAP-based categories to determine the political subcorpus, the clustering procedure and the annotation prompt itself. Yet, this does not make the pipeline a fully automated solution. It still depends on human judgment for those steps that require contextual knowledge and interpretive criteria. This includes labeling clusters from their top terms, deciding when a cluster's vocabulary is ambiguous enough to require closer review, and curating few-shot examples for the specific corpus being analyzed. What this study demonstrates is that scaling conflict framing detection using computational tools does not remove the need for social-science training. It moves the task from coding articles by hand to designing and supervising the pipeline that does.

As for the second question, applying this pipeline at scale revealed that conflict framing is far from uniform across Chilean political coverage. Nevertheless, remaining at the aggregate level is not enough to understand the underlying causal dynamics. The analysis requires disaggregating across multiple levels and incorporating social-scientific context, both of which this study partially addresses, leaving substantial room for future research.

One of the study's broadest findings is that nearly a third of the political subcorpus (29.6%, 215,221 articles) consists of clusters organized around individual or collective actors as protagonists of the narrative, rather than around the theme itself. Within these actor-driven clusters, the conflict frame's weighted mean intensity is more than twice as high (0.41) as in the theme-oriented clusters (0.16). This is likely because this type of structure provides more textual space for conflict to be articulated, since Q1 and Q2 require identifiable parties to anchor to. For this reason, this concentration should not be read as evidence of more political conflict in an absolute sense, but as evidence that the narrative structure. This pattern is closely related to what political communication literature identifies as process- or game-oriented coverage, where the organizing principle of a story is actors in conflict rather than policy substance (Aalberg et al., 2012; Strömbäck, 2008).

Understanding why this pattern emerges requires looking at two structural factors that are not specific to the Chilean case: the kind of content different sources generate, and the incentive

outlets have to highlight that content once it exists. In Government Operations and other topics centered on political actors, such as International Affairs, these sources routinely make adversarial statements as a strategic part of their public role (San Martín Núñez & Guerrero González, 2012). A deputy criticizing a minister is often more a personal attack on the rival's competence or integrity than a substantiated disagreement over policy, a phenomenon that San Martín Núñez and Guerrero González (2012) classify as discrediting. This behaviour already supplies abundant material for the conflict frame's dimensions. However, availability alone does not determine what gets amplified. Since content related to conflict and negativity tends to outperform substantive reporting in audience engagement (Robertson et al., 2023; Khawar et al., 2024), this gives outlets an incentive to select and amplify adversarial exchange whenever it is available.

A similar effect operates in reverse in technocratic areas such as Macroeconomics, Health, or Energy. These domains tend to be organized around a small number of authoritative institutions that generate data-driven, descriptive reporting (Sigal, 1973; Manning, 2001) rather than the competing claims on which conflict framing depends (Semetko and Valkenburg, 2000). Adversarial material is consequently scarcer there to begin with. Even so, whatever conflict-coded coverage does emerge in these domains tends to be amplified accordingly.

This concentration helps to clarify why the study's corpus-wide intensity is roughly half of Gronemeyer and Porath's (2017a) estimate (0.49). The two figures are not directly comparable, since their political subcorpus was restricted to government affairs, party politics and elite actors. Once this study's corpus is narrowed to comparable categories such as Government Operations, the conflict frame intensity rises to 0.42, which is much closer to their reported value. The gap, then, illustrates how much analytical leverage is provided by a broader subcorpus and how much prior estimates may have been shaped by the categories they chose to include.

Crime and immigration are two domains unlocked by this broader subcorpus definition, both worth examining specifically because they have been linked to a documented gap between media coverage and underlying reality in Chile. In the case of crime, it is a particularly instructive case because Law and Crime is the largest category in the corpus by volume, yet one of the lowest in aggregate conflict frame activation. This reinforces the finding that

volume alone does not drive framing. Given that crime coverage in Chile has been linked to a documented gap between media-driven perception and actual victimization (Ajzenman et al., 2023; CEP, 2024), the present findings suggest this gap is unlikely to operate through conflict framing but rather driven by the volume of crime coverage itself. Most crime coverage is narrated from a police or procedural perspective, detentions, homicides, drug seizures, leaving little room for the disagreement or reproach the conflict frame requires. Where activation does concentrate, it follows the same actor-centered pattern identified above: clusters involving high-profile judicial cases, internal security policy, and the Supreme Court.

Immigration follows the same underlying pattern, even though its cluster-level vocabulary appears predominantly administrative (migrants, borders and nationalities). When disaggregated by conflict frame activation, the difference in vocabulary is noticeable: articles where the frame is recognized are marked by named legislators clashing over migration policy, while articles without it are dominated by administrative and police reporting. With 34.9% of its articles carrying a conflict frame, it shows that a substantial share of immigration coverage in Chile is organized around political disputes. This finding complements, from the production side, existing evidence on how media shapes Chilean public attitudes toward migrants (Ajzenman et al., 2023; Scherman et al., 2022), though establishing a direct causal link falls outside the scope of this study.

Having established what drives conflict framing and where it concentrates, a separate question is whether outlets apply it uniformly. Much like the corpus aggregate picture, outlet-level differences in average intensity are not straightforward to interpret. This is because they reflect both editorial choices and each outlet's thematic focus, making it difficult to isolate one from the other. What is clear is that outlets broadly agree on which topics are conflictive while diverging substantially in how intensely they cover them. Parliamentary politics, the single highest-activation cluster in the corpus, is a notable exception since it is covered as confrontational with near-uniform intensity across the ideological spectrum. Identifying why a given outlet covers the same dispute more confrontationally than another would require article-level analysis and more context on the editorial decisions, which falls outside the scope of this study.

Finally, the temporal analysis offers a point of caution against assuming conflict framing scales automatically with political stakes. Among electoral events, the contrast between the

2021 and 2025 presidential elections is particularly telling: the 2021 election shows consistent pre-event elevation across multiple windows, while the 2025 election shows no comparable pattern. What distinguishes the two in terms of conflict frame activation remains an open question this study cannot resolve, though it suggests that formal electoral significance alone is not sufficient to generate conflict-coded coverage.

The *estallido social* also shows elevated activation, though it is harder to interpret in isolation. Rather than reflecting anticipation of a specific unforeseen event, the elevation is consistent with what the broader literature identifies as a longer-term accumulation of institutional and structural tension well underway before October 2019 (Morales Quiroga, 2020; Muñoz-Rodríguez et al., 2024). This study's data can establish the elevation itself, but not the specific causal mechanism behind it.

Beyond the eleven events formally tested, the series shows one additional irregularity worth noting: a sharp and sustained low peak in activation through 2020, coinciding with COVID-19 pandemic and the national quarantine measures. It is consistent with the cluster-level finding reported in Section 4.3.2, where the *pandemia_covid_sanitaria* cluster showed comparatively low conflict-frame activation overall, with the frame appearing only when pandemic coverage turned into political dispute rather than in routine sanitary reporting. Taken together, this suggests that large-scale disruptions to the news agenda such as public health emergencies, wildfires or the death of the ex-president, may depress conflict-frame activation. Two mechanisms could explain this: displacement, where crisis coverage simply takes up the space political reporting that it would otherwise occupy, or a rally effect, where the crisis mutes political division that would normally structure that coverage, as actors and outlets alike converge around the shared emergency. Distinguishing between these explanations would require data this study does not have.

Unlike COVID's one-time disruptions, a second pattern follows a far more predictable cadence. February is the lowest-activation month in every year of the corpus, consistent with Congress being in recess and political activity slowing during the summer holiday period. Also, the two most extreme instances of this seasonal low coincide with major forest fire emergencies, suggesting the dip can be further amplified when disaster coverage crowds out political reporting altogether. At the opposite extreme, July and August are the two highest-activation months, though this peak does not correspond to any specific institutional

event. One plausible explanation is that this period falls furthest from any recess or election-driven disruption, but confirming this would require data this study does not have.

The analysis of conflict framing across topical, outlet-level, and temporal dimensions helps estimating the scale of the phenomenon in Chile and understanding why its implications matter. The most telling indicator is that Government Operations alone accounts for 28.4% of the total political subcorpus, meaning more than one in four political articles is already organized around named institutional figures. Of that coverage, nearly half carries a conflict frame, which means that a substantial portion of what Chilean audiences encounter about their representatives and institutions is coverage through a conflictual lens. The magnitude of this exposure is precisely what makes the democratic implications of conflict framing particularly relevant for the Chilean case. The concentration of conflict framing in political institutions and their named protagonists, documented here, raises a more specific question than the one typically posed in the framing literature. If conflict framing does cultivate political cynicism, as Cappella and Jamieson (1997) argue, this concentration suggests that cynicism may fall disproportionately on the formal political actors who dominate conflict-coded coverage. Citizens' relationship with technocratic institutions, by contrast, may remain comparatively unaffected. This distinction cannot be resolved with the production-side data presented here

Future lines of research could expand on both the pipeline and the substantive findings. Methodologically, the pipeline could be applied to other languages and media systems to test whether the actor-centered pattern documented here is a general feature of conflict framing in political journalism or specific to the Chilean case. On the substantive side, future work could dive deeper into any of the dynamics identified such as outlet-level editorial decisions, the temporal drivers that distinguished the 2021 from the 2025 election, or pursue the audience-side research needed to test the hypothesis this study raises.

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7. Appendices

Appendix 1: Pipeline Hyperparameters

Stage	Parameter	Value
TF-IDF Vectorization	Maximum vocabulary size (max_features)	100,000
	Minimum document frequency (min_df)	10
	Term frequency normalization	Sublinear (log)
	N-gram range	Unigrams (1,1)
	Stopwords	Spanish (spaCy ES stopword list)
	Token pattern	Words of 3+ characters, Spanish alphabet
LSA (TruncatedSVD)	Dimensions (n_components)	500
	SVD iterations (n_iter)	10
	Algorithm	Randomized
	Post-SVD normalization	L2 (enables cosine similarity via dot product)
MiniBatch K-Means	Batch size	4,096
	Initializations (n_init)	10
	Random state	42
	K values evaluated	{20, 30, 40, 50, 60, 70, 80}
Silhouette Evaluation	Stratified sample size	10,000
	Distance metric	Cosine
	Marginal improvement threshold	$\Delta < 0.001$

Appendix 2: Clusters

Cluster ID	Nº docs	Top Words	Cluster Label	CAP Topic Code	CAP Topic Name
2	31,333	salud, covid, cuarentena, coronavirus, casos, pandemia, sanitaria, personas, medidas, virus, contagios, región, seremi, autoridad, país	pandemia_covid_sanitaria	3	Health
3	22,141	fiscalía, fiscal, investigación, caso, público, ministerio, abogado, delitos, regional, hermosilla, querrella, causa, antecedentes, delito, defensa	casos_judiciales_alto_perfil	12	Law and Crime
4	18,568	carabineros, detenidos, policial, personas, personal, detención, lugar, robo, comisaría, vehículo, comuna, sujetos, santiago, control, sector	carabineros_detenciones	16	Law and Crime
5	11,342	convención, constitucional, constitución, constituyente, convencionales, constituyentes, propuesta, comisión, proceso, pueblos, órgano, convencional, chile, pleno, derecho	convencion_constitucional	20	Government Operations
6	23,924	empresa, millones, mercado, empresas, compañía, chile, inversión, desarrollo, año, industria, país,	empresas_inversion	15	Domestic Commerce

		años, firma, mundo, proyectos			
7	14,326	hombre, víctima, disparos, homicidios, hospital, pdi, brigada, lugar, comuna, carabineros, homicidio, años, vital, bala, ocurrió	homicidios_pdi	12	Law and Crime
8	6,868	dólar, cobre, mercado, mercados, alza, analista, billete, precio, unidos, caída, china, comprador, verde, xtb, latam	mercados_financieros	15	Banking
10	15,367	constitución, plebiscito, constitucional, proceso, rechazo, apruebo, constituyente, convención, propuesta, partidos, chile, partido, presidente, opción, texto	plebiscito_apruebo_rechazo	20	Government Operations
12	7,363	metro, estaciones, estación, línea, santiago, tren, servicio, trenes, red, hrs, transporte, pasajeros, disponible, subterráneo, baquedano	metro_transportesantiago	10	Transportation
13	18,889	corte, suprema, fallo, tribunal, apelaciones, recurso, judicial, sentencia, resolución, máximo, sala, caso, protección, decisión, santiago	poder_judicial_corte_suprema	12	Law and Crime
14	13,868	trabajadores, trabajo, empresa, laboral, sindicato, huelga, gobierno, paro, laborales, trabajadoras, empresas, salud, nacional, trabajador, movilización	trabajadores_sindicatos	5	Labor and Employment

15	19,430	seguridad, interior, gobierno, carabineros, ministra, monsalve, tohá, crimen, subsecretario, violencia, organizado, país, presidente, policías, delitos	seguridad_politica_interior	16	Law and Crime
16	5,007	ejército, militar, corte, defensa, jefe, rutherford, comandante, general, ministra, libertad, institución, fraude, apelaciones, recurso, romy	ejercito_ffaa	16	Defense
17	17,591	productos, sernac, consumidores, precios, salud, precio, servicio, empresa, consumidor, nacional, usuarios, mercado, clientes, empresas, producto	sernac_consumidores	15	Domestic Commerce
18	25,487	partido, elecciones, candidato, presidencial, candidatura, partidos, electoral, candidatos, primarias, política, elección, democracia, chile, candidata, frente	partidos_elecciones	20	Government Operations
19	10,182	reforma, pensiones, gobierno, previsional, proyecto, tributaria, sistema, hacienda, oposición, afp, trabajo, presidente, comisión, pensión, marcel	reforma_previsional_congreso	13	Social Welfare
21	7,987	gendarmería, cárcel, penal, penitenciario, internos, sexual, recinto, funcionarios, denuncia, abuso, justicia, institución, interior, reos, centro	gendarmeria_carceles	12	Law and Crime

23	21,719	gobierno, ministro, presidente, ministra, boric, vallejo, cargo, gabinete, gabriel, ministerio, ejecutivo, hacienda, camila, política, renuncia	voceria_gabinete	20	Government Operations
24	10,510	casos, covid, positividad, salud, contagios, minsal, pcr, fallecidos, camas, pacientes, activos, exámenes, coronavirus, pandemia, decesos	covid_cifras_minsal	3	Health
25	2,871	precios, ipc, índice, inflación, variación, ine, mensual, alza, mes, meses, consumidor, estadísticas, divisiones, alimentos, instituto	ipc_inflacion_estadisticas	1	Macroeconomics
26	15,947	obras, construcción, proyecto, vivienda, viviendas, públicas, millones, familias, mil, región, mop, ministerio, inversión, regional, sector	vivienda_minvu	14	Housing
27	8,114	carabinero, carabineros, funcionario, arma, policial, vehículo, herido, sujeto, fuego, hospital, disparos, lugar, uniformado, comuna, policía	carabineros_incidentes	16	Law and Crime
28	9,134	pago, bono, beneficio, monto, mil, subsidio, familiar, personas, millones, familias, ingresos, social, hogares, requisitos, beneficios	bonos_subsidios_sociales	13	Social Welfare
29	10,719	contraloría, república, general, informe, irregularidades, funcionarios, contralor,	contraloria_municipios	20	Government Operations

		públicos, millones, recursos, sumario, organismo, municipalidad, municipio, licencias			
31	8,081	juicio, oral, tribunal, fiscalía, penal, audiencia, acusado, homicidio, caso, años, cárcel, defensa, presidio, fiscal, público	juicios_penales	12	Law and Crime
32	9,706	año, trimestre, millones, aumento, alza, crecimiento, meses, caída, mes, tasa, cifras, cifra, baja, comercio, ventas	indicadores_economicos	1	Macroeconomics
33	22,136	universidad, chile, estudio, mundo, años, desarrollo, investigación, estudios, personas, salud, vida, forma, educación, trabajo, país	universidad_investigacion	17	Education
34	9,073	retiro, fondos, afp, pensiones, proyecto, previsionales, gobierno, diputados, ahorros, iniciativa, cámara, retiros, afiliados, comisión, reforma	retiros_fondos_afp	13	Social Welfare
35	10,875	prisión, preventiva, arresto, domiciliario, cautelar, garantía, juzgado, medida, imputado, imputados, audiencia, tribunal, fiscalía, delitos, investigación	prision_preventiva	12	Law and Crime
36	8,653	matthei, evelyn, providencia, kast, candidata, presidencial, alcaldesa, kaiser, chile,	candidatos_presidenciales	20	Government Operations

		jara, antonio, udi, partido, josé, johannes			
37	19,060	boric, presidente, gabriel, mandatario, chile, gobierno, relaciones, país, países, exteriores, canciller, jefe, república, política, argentina	gobierno_boric	19	International Affairs
39	7,836	dosis, vacunación, vacuna, vacunas, salud, covid, pfizer, sinovac, población, personas, refuerzo, inoculación, virus, minsal, paris	vacunacion_covid	3	Health
40	10,834	mujeres, género, mujer, violencia, hombres, equidad, sexual, ministra, derechos, ley, aborto, chile, feminista, orellana, años	genero_violencia_mujer	2	Civil Rights
42	19,147	diputados, diputado, cámara, acusación, parlamentarios, partido, comisión, udi, diputada, constitucional, bancada, oposición, presidente, parlamentario, congreso	politica_parlamentaria_interna	20	Government Operations
43	13,703	kast, antonio, josé, republicano, candidato, partido, presidencial, boric, presidente, vuelta, chile, gobierno, jara, electo, campaña	candidatura_gobierno_kast	20	Government Operations
44	16,696	derechos, humanos, violaciones, dictadura, indh, chile, justicia, gobierno, víctimas, social, país, nacional, estallido, carabineros, personas	ddhh_dictadura_memoria	2	Civil Rights

46	9,410	drogas, droga, cocaína, tráfico, pdi, marihuana, detenidos, operativo, kilos, banda, detención, armas, criminal, investigación, policía	narcotrafico_dr ogas	12	Law and Crime
47	20,146	proyecto, ley, senado, cámara, comisión, diputados, iniciativa, sala, aprobó, gobierno, votos, congreso, favor, busca, ejecutivo	legislativo_proy ectos_ley	20	Government Operations
48	4,199	israel, gaza, israelí, hamás, palestina, franja, palestino, israelíes, palestinos, guerra, ataques, netanyahu, internacional, conflicto, ataque	conflicto_israel _gaza	19	International Affairs
51	13,099	transporte, vehículos, transportes, buses, tránsito, santiago, conductores, ruta, región, público, metropolitana, mil, pasajeros, horas, vehicular	transporte_buse s	10	Transportation
52	7,731	mapuche, pueblo, pueblos, comunidades, araucanía, indígenas, indígena, comunidad, originarios, tierras, catrillanca, gobierno, territorio, comunero, mapuches	araucania_confl icto_mapuche	2	Civil Rights
54	15,698	robo, delincuentes, vehículo, sujetos, carabineros, asalto, víctima, comuna, auto, armas, lugar, antisociales, fuego, violento, especies	robos_delincue ncia	12	Law and Crime

55	11,236	prisión, preventiva, imputado, víctima, hombre, acusado, homicidio, investigación, mujer, fiscalía, garantía, fiscal, delito, años, juzgado	imputados_homicidio	12	Law and Crime
56	12,675	trump, unidos, donald, estadounidense, presidente, guerra, aranceles, china, blanca, países, estadounidenses, ucrania, rusia, washington, país	politica_internacional_trump	19	International Affairs
57	20,918	alcalde, municipio, municipalidad, municipal, comuna, vecinos, comunal, alcaldesa, municipales, seguridad, concejo, santiago, ciudad, jefe, personas	municipios_alcaldes	20	Government Operations
58	8,132	migrantes, extranjeros, país, chile, migratoria, frontera, migración, personas, gobierno, venezolanos, venezuela, irregular, ciudadanos, Perú, bolivia	migrantes_venezuela	9	Immigration
59	10,096	economía, banco, crecimiento, inflación, central, año, pib, tasa, monetaria, mercado, política, tasas, económico, precios, interés	economia_banco_central	1	Macroeconomics
60	4,962	iglesia, papa, abusos, francisco, obispo, sexuales, sacerdote, abuso, católica, vaticano, karadima, sexual, víctimas, obispos, fernando	iglesia_abusos_sexuales	2	Civil Rights

61	14,102	agua, suministro, empresa, potable, aguas, clientes, eléctrico, región, corte, zona, luz, servicio, comunas, enel, superintendencia	servicios_basicos_agua_luz	8	Energy
63	9,656	presidio, años, pena, tribunal, cárcel, condenado, condena, oral, sentencia, delito, penal, juicio, fallo, condenó, delitos	condenas_juicios	12	Law and Crime
64	7,564	ataque, araucanía, incendiario, arauco, región, sur, carabineros, violencia, mapuche, ruta, malleco, lugar, provincia, zona, camiones	araucania_atques_incendarios	16	Law and Crime
65	18,478	educación, estudiantes, clases, colegio, profesores, escolar, establecimientos, alumnos, establecimiento, colegios, liceo, apoderados, mineduc, niños, educacionales	educacion_escolar	17	Education
67	15,867	piñera, sebastián, presidente, mandatario, gobierno, chile, país, moneda, ministro, expresidente, social, república, jefe, chilenos, nacional	gobierno_pinera	20	Government Operations
68	14,353	detenido, detención, hombre, sujeto, carabineros, pdi, años, policía, robo, control, delito, detienen, acusado, policial, imputado	detenidos_policial	12	Law and Crime

69	11,691	cuerpo, pdi, brigada, homicidios, hallazgo, hombre, muerte, policía, cadáver, encontrado, investigaciones, diligencias, víctima, lugar, médico	hallazgos_homicidios_pdi	12	Law and Crime
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Table A2. Cluster labels, top terms, and corresponding CAP topics

Appendix 3: Event Window Percentiles

Analysis of the percentile shifts in the binary conflict frame indicator (C1 OR C2) around each of the eleven political events. For pre-event windows of 1, 3, and 6 months are defined while for post-events, windows of 1 month. Δ is the difference between the window's mean activation rate and the series' overall mean, in percentage points. %ile is the percentile of that Δ within the null distribution of all equivalent-length windows across the series.

Event	Period	Pre 1m		Pre 3m		Pre 6m		Post 1m	
		Δ	%ile	Δ	%ile	Δ	%ile	Δ	%ile
Inicio gobierno Piñera 2	2018-03	1.71	70.7	4.58	94.8*	4.58	100.0*	6.81	98.0
Estallido social	2019-10	4.35	91.9	4.02	91.8	4.17	96.8	3.01	79.8
Plebiscito Constitución	2020-10	2.92	77.8	0.96	66.0	-2.39	13.8	2.87	75.8
Elección presidencial 1ra vuelta	2021-11	4.31	91.9	2.44	77.3	0.59	69.1	2.55	74.7
Elección presidencial 2da vuelta	2021-12	4.35	91.9	4.19	92.8	1.64	75.5	-3.33	15.2
Inicio gobierno Boric	2022-03	-3.89	9.1	-1.56	28.9	1.31	73.4	-0.40	46.5
Plebiscito de salida	2022-09	4.12	88.9	0.36	55.7	-0.73	45.7	-2.59	21.2
Plebiscito Constitución 2	2023-12	-3.45	12.1	-2.56	16.5	-1.17	36.2	1.11	64.6
Muerte Piñera / Incendios	2024-02	1.11	64.6	-1.30	36.1	-1.11	37.2	-1.16	37.4
Elección presidencial 1ra vuelta (Kast)	2025-11	1.09	63.6	0.42	56.7	-0.25	56.4	-0.59	43.4
Elección presidencial 2da vuelta (Kast)	2025-12	0.96	61.6	0.75	63.9	0.28	67.0	-1.04	39.4

* Window falls partially or fully outside the corpus period (starts January 2018) and is based on fewer than the full number of months.

Appendix 4 — Code and Corpus

The source code and full dataset used for corpus construction, TF-IDF/LSA vectorization, clustering, LLM-based conflict frame annotation, and downstream analysis are available at the following repository:

Link: <https://drive.google.com/drive/u/2/folders/102eibmrJWe-6auv8efbpZ-cLelasfbsw>

The repository includes:

- **Methodology/**

Jupyter notebooks covering the full pipeline: corpus construction (1_construir_corpus.ipynb), TF-IDF vectorization and clustering (2.tfidf_clustering.ipynb), cluster definitions and CAP topic mapping (clusters_definition.xlsx), CAP-based political subcorpus generation (generar_corpus_cap.ipynb), pilot testing of the annotation prompt (pilot_cluster_x_frame.ipynb), full-scale QWEN3-8B annotation of the 734,500-article political subcorpus (anotar_cap_734k.ipynb), the discarded XLM-RoBERTa fine-tuning experiment (finetune_silver.ipynb), the event-window percentile analysis (event_window_percentiles.ipynb), and figure generation for the paper (generate_paper_figures.ipynb)

- **Human annotations/**

the gold-standard human annotations used to validate the LLM pipeline (the 296-article agreement sample and the 146-article held-out gold set referenced in Section 4.1)

- **Corpus/**

the raw scraped corpus (prensa_datos.parquet, scrapingbiobio.csv), the cleaned corpus (corpus_limpio_final_v3.parquet), the clustering output (final_clustering_k70.parquet), the CAP-filtered political subcorpus (corpus_cap_734k.parquet), and the final fully annotated corpus (corpus_completo_annotado.parquet)

- **paper_figures/**

the figures included in this report

Given the size of the dataset (~7.5 GB, exceeding both direct attachment and Git LFS free-tier limits), it was hosted on Google Drive to ensure traceability and reproducibility of the analysis described in Chapter 3 (Methodology).